WEEK 4 / LESSON 2 / 50 MINUTES

L2

Computer Vision

DigiYatra at airport gates. Phone unlock. Office attendance systems. Face detection is the single most deployed CV task in India — and today, you build it.

Which airport gate did you last face-match at? What happened if it failed?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

Aadhaar authenticates 1.3 billion faces.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. The 4-stage CV pipeline — detect, align, embed, compare.
2. Run MTCNN for face detection + landmarks.
3. Generate FaceNet 128-dim embeddings.
4. Compare faces with cosine similarity.
5. Build a face recogniser in 80 lines.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
The 4-stage pipeline
Detection finds WHERE. Embedding turns it into a FINGERPRINT. Compare says same-or-different.

02
MTCNN — detection +...
Multi-task CNN

03
FaceNet — 128 dimensions of...
Embedding, not classification

04
Cosine similarity — the...
Typical thresholds

01

PART 1 OF 4

The 4-stage pipeline

Detection finds WHERE. Embedding turns it into a FINGERPRINT. Compare says same-or-different.

CONCEPT 1 OF 4

The 4-stage pipeline

Detection finds WHERE. Embedding turns it into a FINGERPRINT. Compare says same-or-different.

DEFINITION

The 4-stage pipeline

Detection finds WHERE. Embedding turns it into a FINGERPRINT. Compare says same-or-different.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

MTCNN — detection + landmarks

Three cascaded CNNs: P-Net, R-Net, O-Net.
Outputs bounding boxes + 5 landmarks (eyes, nose, mouth corners).
Landmarks enable alignment — huge accuracy boost.
facenet-pytorch wraps it in 3 lines.
OUTPUTS
Boxes + landmarks

Downstream stages depend on both.

KEY POINTS

- Three cascaded CNNs: P-Net, R-Net, O-Net
- Outputs bounding boxes + 5 landmarks (eyes, nose, mouth corners)
- Landmarks enable alignment — huge accuracy boost
- facenet-pytorch wraps it in 3 lines

CONCEPT 3 OF 4

FaceNet — 128 dimensions of identity

160×160 face crop → 128-dim unit vector.
Trained with triplet loss.
Same person → close in cosine space.
Add a new person to database? Just embed them once.
TRICK
Embed, don't classify

No retraining for new people.

DEFINITION

FaceNet — 128 dimensions of identity

Embedding, not classification

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Cosine similarity — the decision

sim > 0.7 → same person (tune to your data).
0.4 < sim < 0.7 → uncertain (ask human).
sim < 0.4 → different person.
Always calibrate on your own gallery before deploy.
CALIBRATE
Tune per dataset

Thresholds drift with camera, lighting, age.

KEY POINTS

- sim > 0.7 → same person (tune to your data)
- 0.4 < sim < 0.7 → uncertain (ask human)
- sim < 0.4 → different person
- Always calibrate on your own gallery before deploy

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Calibrate your threshold

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.

2. Calibrate your threshold

3.

4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1	*Where would a 5% false-reject be unacceptable?*
Q2	*Where would a 5% false-accept be catastrophic?*
Q3	*What privacy controls would you add before deploying this at a college gate?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Stage between detect and embed?

A Filter

B Align

C Classify

D Sort

ANSWER · QUESTION 1 OF 3

Stage between detect and embed?

CORRECT ANSWER

B

Align

WHY

Alignment using MTCNN landmarks standardises pose before embedding

KNOWLEDGE CHECK · QUESTION 2 OF 3

FaceNet outputs:

A Label

B Bounding box

C 128-dim vector

D Mask

ANSWER · QUESTION 2 OF 3

FaceNet outputs:

CORRECT ANSWER

C

128-dim vector

WHY

A unit-norm 128-dim embedding

KNOWLEDGE CHECK · QUESTION 3 OF 3

Why no retraining for new people?

A Model is frozen

B It embeds, not classifies

C Gallery is small

D Speed

ANSWER · QUESTION 3 OF 3

Why no retraining for new people?

CORRECT ANSWER

B

It embeds, not classifies

WHY

Add a new person = add one embedding to the gallery. No weight updates

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Detection finds WHERE. Embedding turns it into a FINGERPRINT. Compare says same-or-different.

THE TAKEAWAY

"That is a decade of research compressed to a heuristic you can type in 80 lines."

WHAT TO NOTICE

1. The 4-stage pipeline
2. MTCNN — detection +...
3. FaceNet — 128 dimensions of...
4. Cosine similarity — the...

APPLY IT → ASSIGNMENT

Mini assignment — family recogniser

Build a family recogniser using 3 photos each of 3 family members.

DELIVERABLES

- Compute gallery embeddings offline.
- Run webcam demo; label faces in real time.
- Record a 30-second screen capture.
- Upload to GitHub + post link.

WHAT 'GOOD' LOOKS LIKE

Deliverable: GitHub URL + 30-second demo video.

ONE THING TO REMEMBER

"That is a decade of research compressed to a heuristic you can type in 80 lines."

— AI Learning Hub

END OF LESSON 4

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 3 — Transformers. The architecture behind every modern LLM.